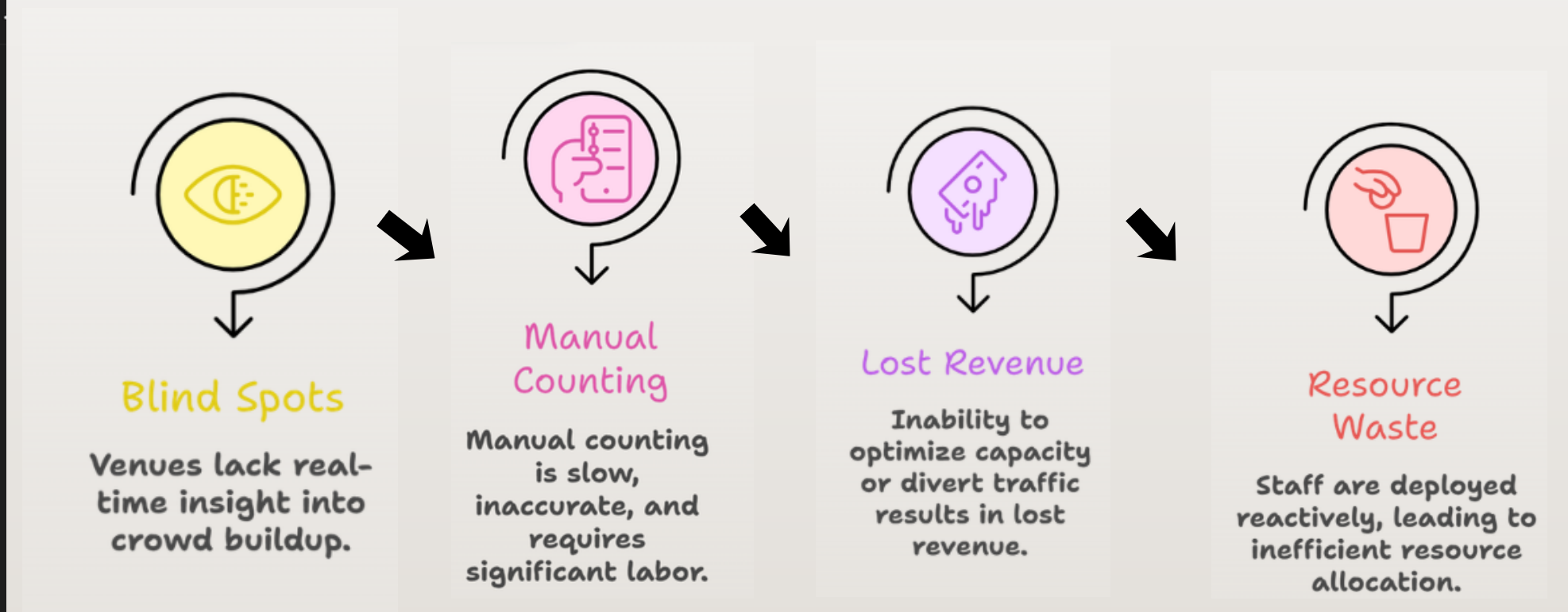
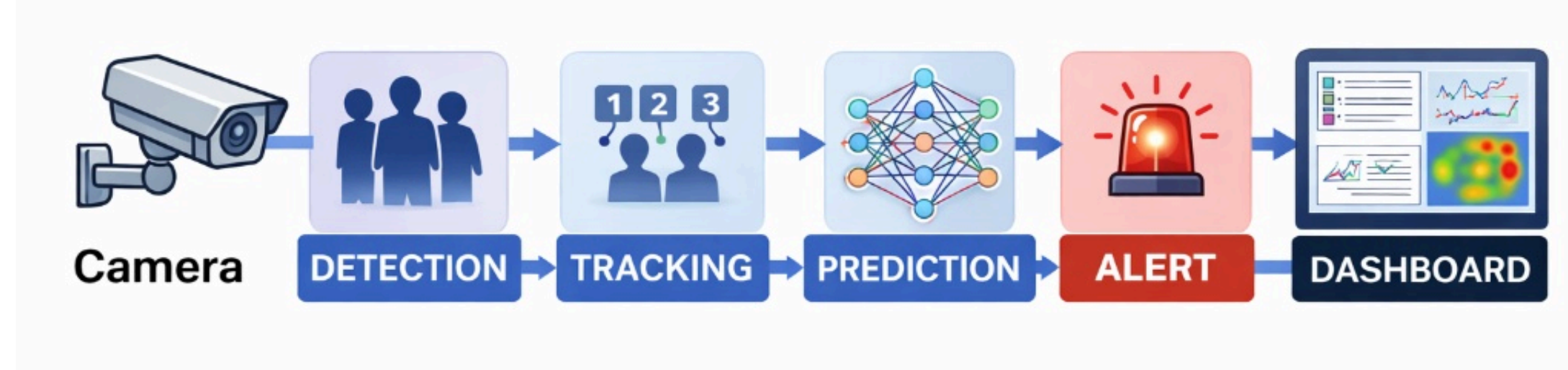




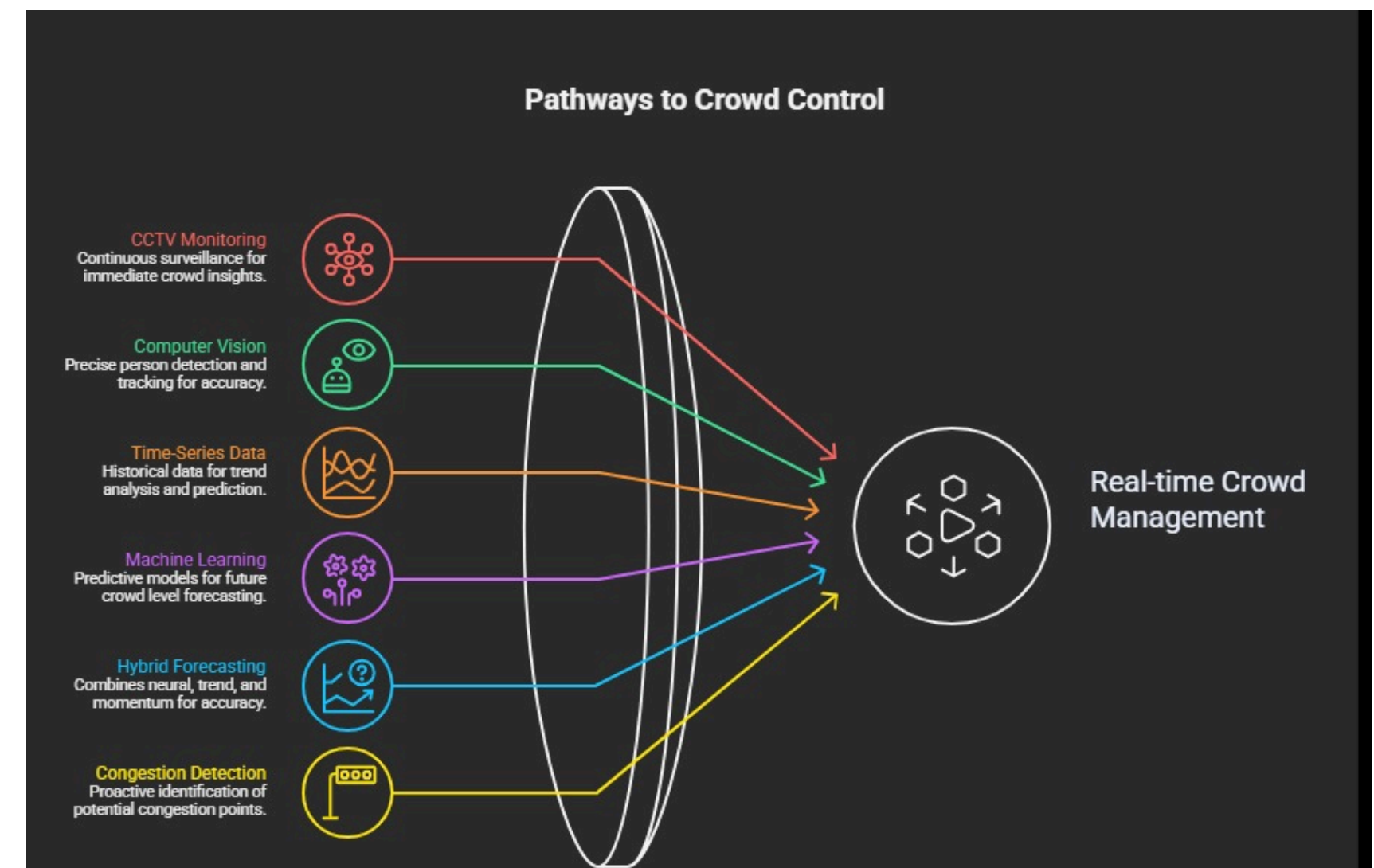
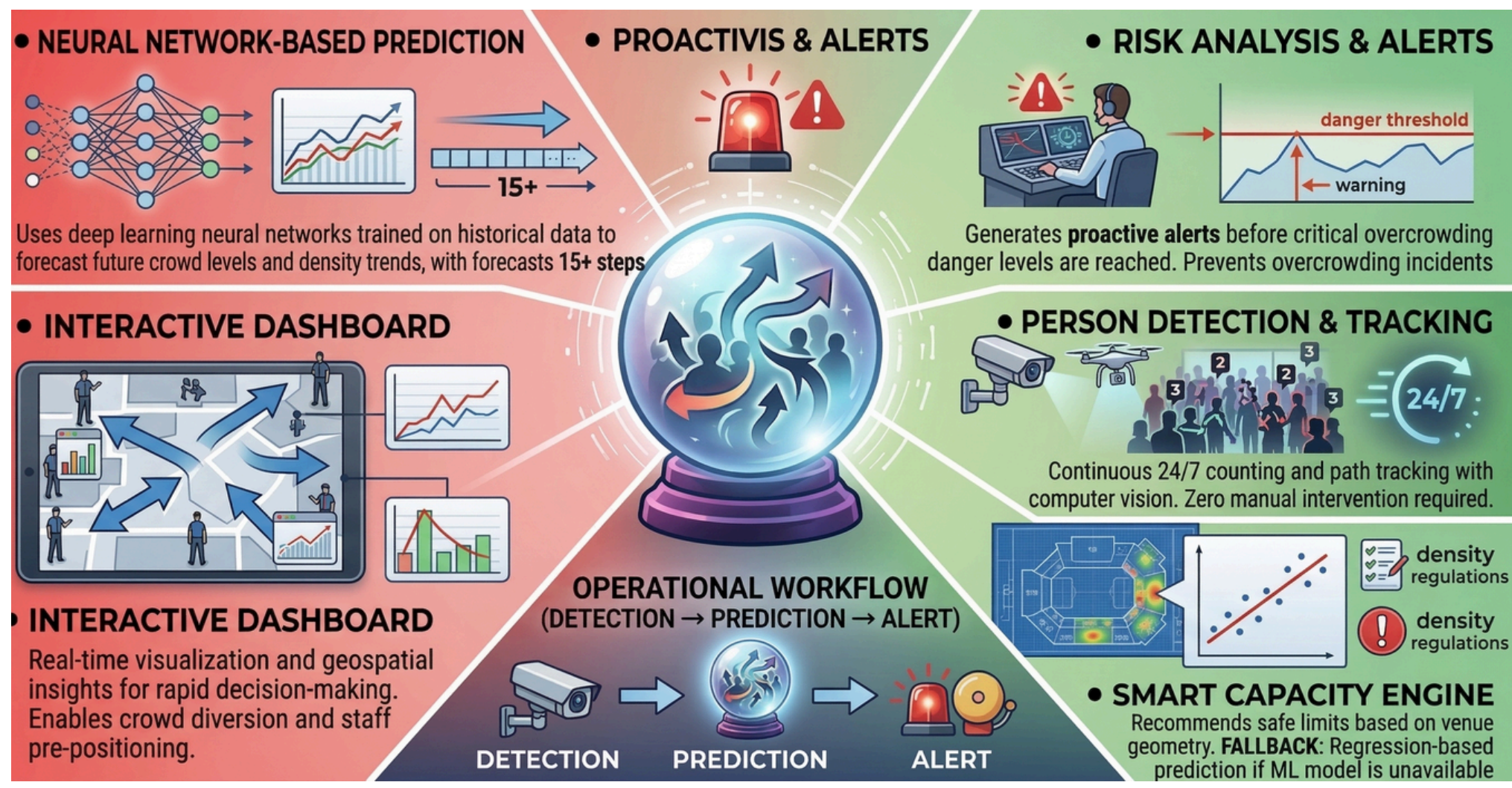
PROBLEM CONTEXT

- ✦ Develop an autonomous crowd density prediction system using computer vision and AI to monitor public spaces and provide early warnings of overcrowding or potential stampede risks.
- ✦ Large-scale public gatherings experience unpredictable crowd flow leading to sudden congestion, while existing CCTV-based monitoring lacks future forecasting-time analytics. This limits visibility into crowd movement and future buildup, making decision-making reactive and preventing early warnings of overcrowding or potential stampede risks.





We propose an AI-driven framework that detects, tracks, and predicts crowd dynamics in real time by combining computer vision, temporal modeling, and risk-aware analytics. From detection to early warning it monitors current crowd, predict near-future surge, and trigger actionable alerts.

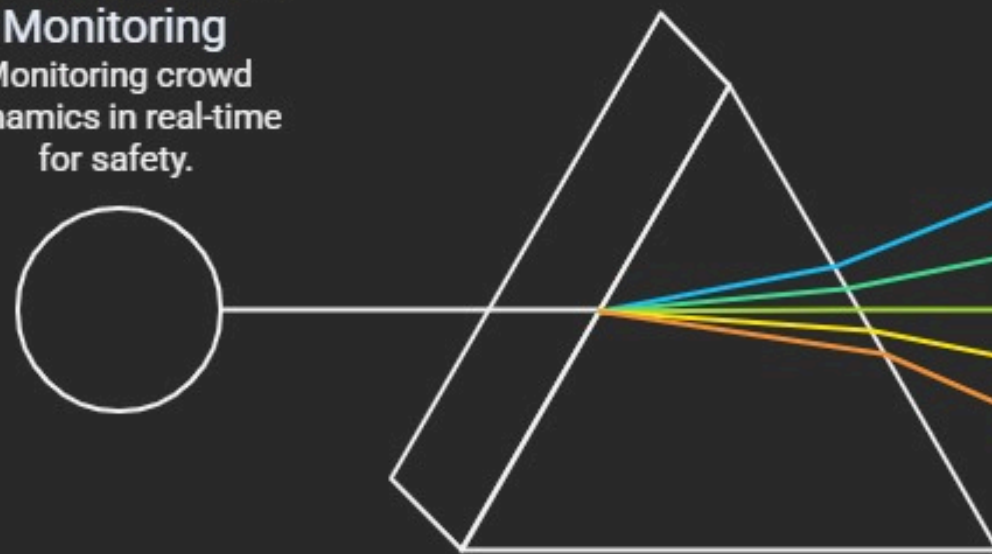


IDEA APPROACH

- Build an intelligent system that not only detects current crowd density but also predicts near-future crowd behavior.
- Combine computer vision, tracking, and neural forecasting to generate early warning signals.
- Convert raw predictions into actionable risk levels and operator recommendations.
- Our model Acheived Acuuracy around 72 % , which currently many real world systems failed to ensure it

Unveiling the Dimensions of Real-Time Crowd Monitoring

Real-Time Crowd Monitoring
Monitoring crowd dynamics in real-time for safety.



- Accurate Person Detection**
Identifying individuals with high precision in video feeds.
- Unique Identity Continuity**
Tracking individuals consistently across multiple video frames.
- Predictive Crowd Count**
Forecasting future crowd numbers to anticipate needs.
- Reduced False Alarms**
Minimizing errors in crowd detection using logic.
- Data-Driven Safety Decisions**
Enabling informed choices for enhanced crowd safety.

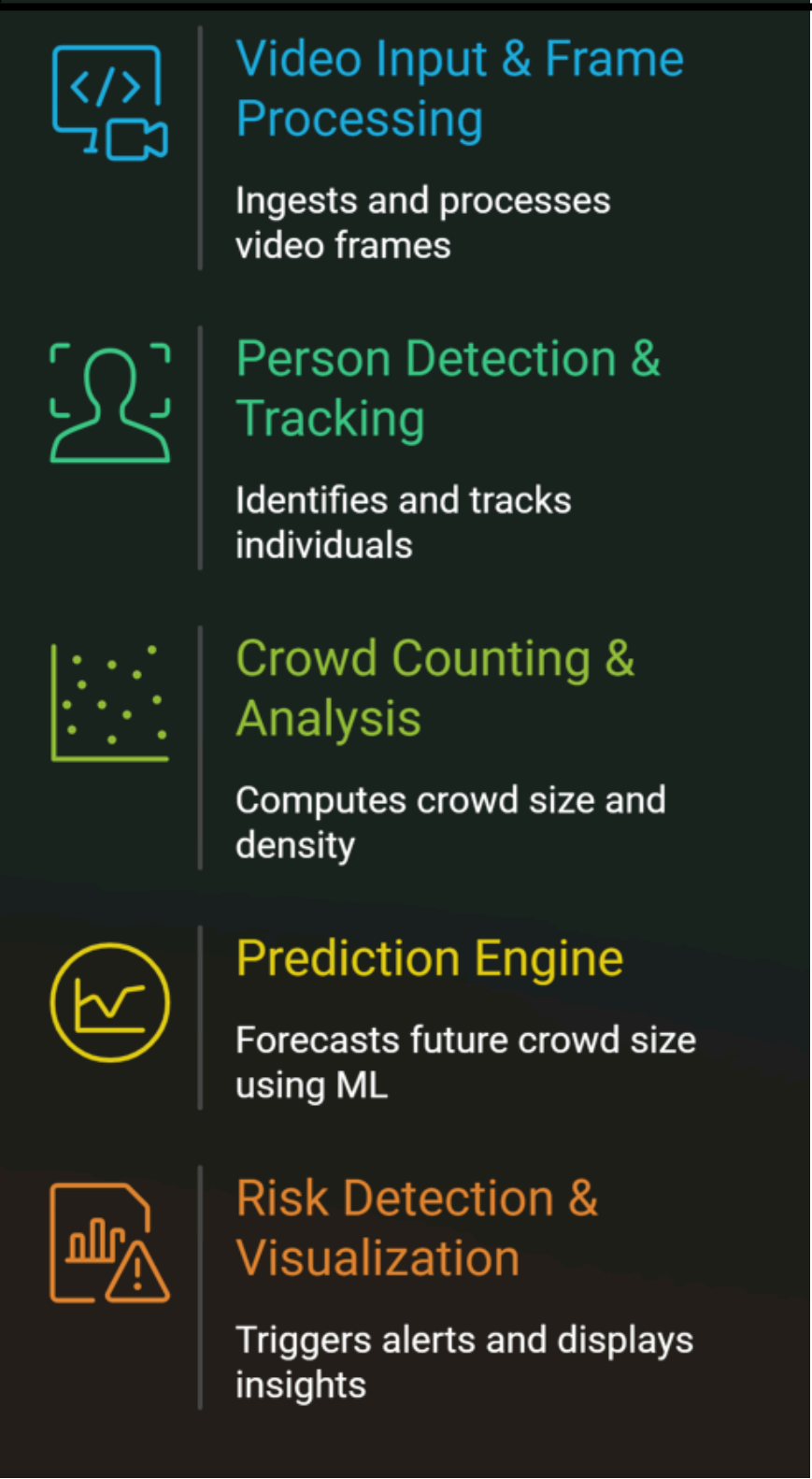
AI-Powered CROWD PREDICTION SYSTEM



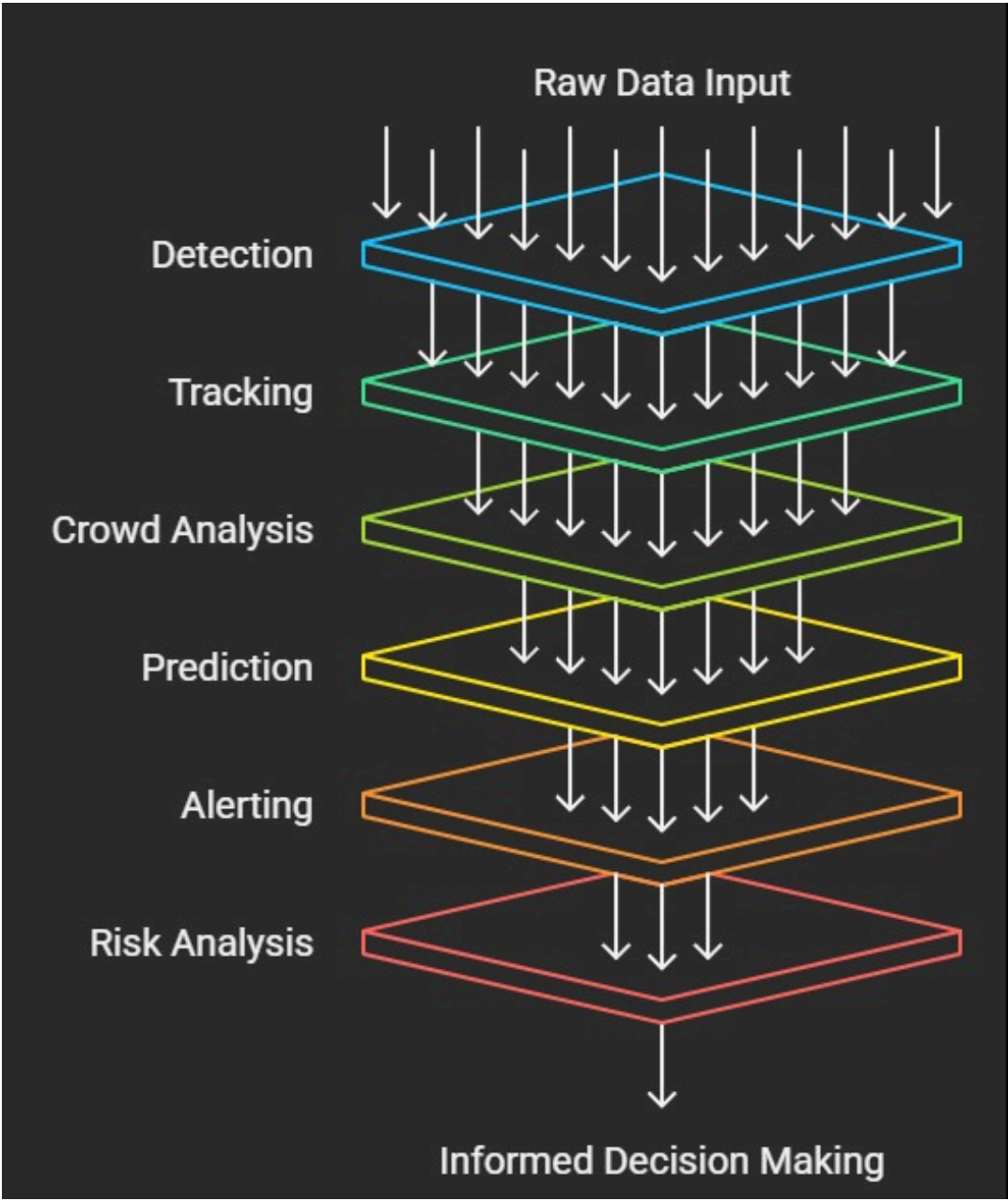
SYSTEM ARCHITECTURE



WORKFLOW

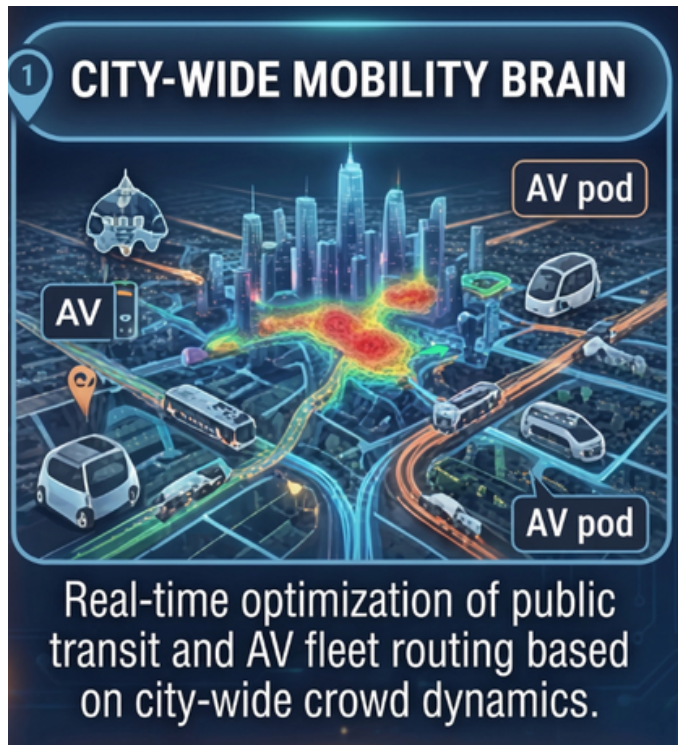


LAYERS OF Cognivue AI



FUTURE SCOPE

1 CITY-WIDE MOBILITY BRAIN



Real-time optimization of public transit and AV fleet routing based on city-wide crowd dynamics.

Deep Learning Models

Integrate LSTM/Transformer-based models for more accurate long-term crowd prediction.

Edge AI Deployment

Run models on edge devices (cameras, local servers) to reduce latency and dependency on cloud.

IoT Integration

Connect with sensors, traffic systems, and public announcement systems for automated control.

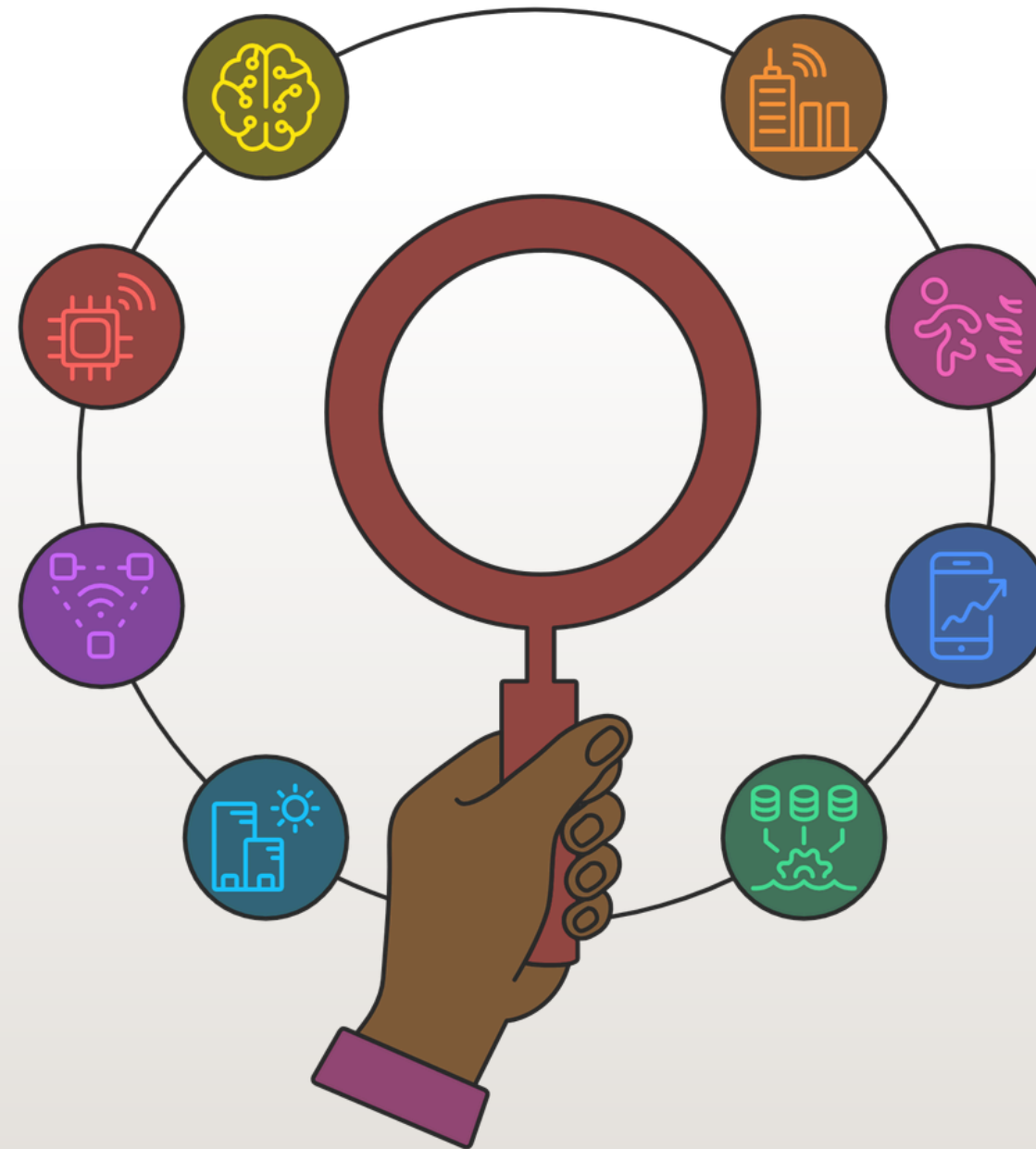
Predictive Simulation

Simulate crowd scenarios in advance for better planning of events and emergencies.

AR DIGITAL TWIN OPTIMIZATION



Designing sustainable and resilient public spaces based on long-term crowd density predictions.



City-Scale Integration

Connect multiple CCTV feeds across locations for smart city-level monitoring.

Behavior Analysis

Detect unusual activities like panic movement, stampede patterns, or sudden dispersal.

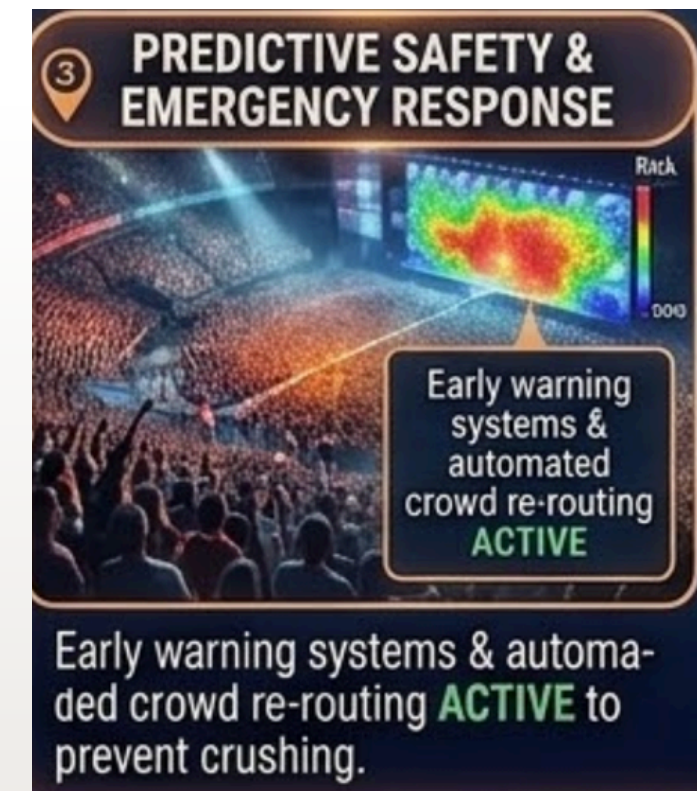
Mobile App

Real-time alerts, heatmaps, and decision support on handheld devices for authorities.

Real-World Data

Continuous learning using live data to enhance model performance over time.

3 PREDICTIVE SAFETY & EMERGENCY RESPONSE



Early warning systems & automated crowd re-routing **ACTIVE** to prevent crushing.

BUSINESS SALE POINT OF VIEW

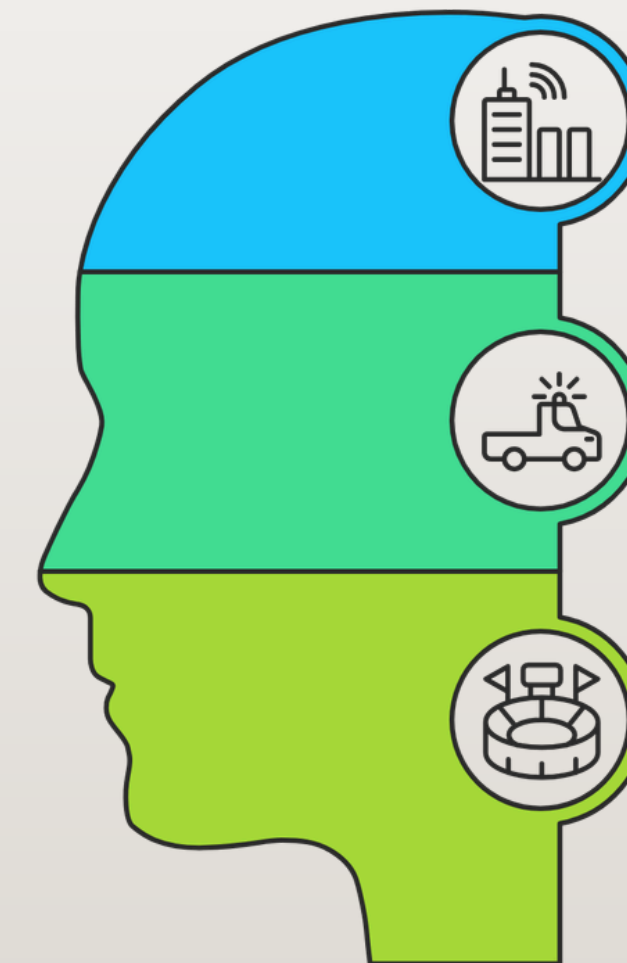
**“I CONVERT NOW INSPECT THE VIDEO INTO ACTION:
DETECT NOW, PREDICT NEXT, INTERVENE EARLY.”**



Business Model



Target Audience



Government & Smart City Authorities

Enhances public safety through large-scale monitoring

Police & Disaster Management Teams

Facilitates rapid response to high-risk crowd situations

Event Organizers & Stadium Operators

Ensures safe crowd flow during large gatherings

demo clips of my proj

Crowd Timeline Analyzer

Recommendation: NORMAL_MONITORING

Risk: SAFE | Congestion: 2.00

Gate=DELTA_BELOW_THRESHOLD, trend=STABLE, flow=UP, congestion=2.0, action=NORMAL

Future Crowd Prediction

Now: 18 Future: 17 Delta: -1

Confidence: 69.0% (MEDIUM) Incoming Prob: 44.3%

Current Crowd Tracking -> Observed: 18 | Predicted Current: 18

Model State: NN_ONLY_DEMO | Gate: DELTA_BELOW_THRESHOLD

Action: NORMAL | Streak: 0 | Surge: NO

Future Risk Status: SAFE | Horizon: 15 frames (~0.50s)

Elapsed: 11.0% | Predicted increase below alert threshold

Predict @ frame 222: now 18 -> future 17 (confidence 69.0%, nowcast 18, horizon 15 frames (~0.50s)) | NORMAL_MONITORING

Keys: r=Run s=Stop p=Predict Studio o=Overlay a=AutoPredict c=Reboot q=Quit

Run Stop Predict Studio Overlay: ON AutoPredict: ON Reboot Timeline

Crowd Timeline Analyzer

Recommendation: NORMAL_MONITORING

Risk: SAFE | Congestion: 3.00

Gate=BEFORE_10_PERCENT, trend=STABLE, flow=RIGHT, congestion=3.0, action=NORMAL

Future Crowd Prediction

Status: Waiting for baseline analysis

Prediction unlocks at 10% timeline

Current progress: 5.4% | Remaining: 4.6%

Predict Studio is enabled only after analysis threshold

Main2 Person ID Preview

Unique IDs: 14

ID: 2 ID: 7

ID: 3 ID: 4

Page 3/4

Predict @ frame 110: now 14 -> future 14 (confidence 69.2%, nowcast 14, horizon 15 frames (~0.50s)) | NORMAL_MONITORING

Keys: r=Run s=Stop p=Predict Studio o=Overlay a=AutoPredict c=Reboot q=Quit

Run Stop Predict Studio Overlay: ON AutoPredict: ON Reboot Timeline

Future Crowd Prediction

Now: 14 Future: 15 Delta: +1

Confidence: 69.0% (MEDIUM) Incoming Prob: 55.7%

Current Crowd Tracking -> Observed: 14 | Predicted Current: 13

Model State: NN_ONLY_DEMO | Gate: DELTA_BELOW_THRESHOLD

Action: NORMAL | Streak: 0 | Surge: NO

Future Risk Status: SAFE | Horizon: 15 frames (~0.50s)

Studio @33%. 0 (-10) risk SAFE | surge NO

Prediction Timeline Studio

Future %: 21

PREDICTION TIMELINE STUDIO

Model output uses trained crowd predictor + observed current—video history. No synthetic metadata is injected.

Current timeline: 19.9% Target timeline: 21% (frame 425/2020)

Future horizon: 22 frames (~0.73s)

Now Crowd: 14 Current Nowcast: 14

Predicted Crowd: 16 Confidence: 69.3% (MEDIUM)

Delta: +2 Incoming Prob: 61.3%

Future Crowd Surge: NO Future Risk: SAFE

Predicted increase below alert threshold Press Enter to apply this prediction to main screen

Esc/Q: Close Simulation checkpoints -> 19%:15 | 20%:14 | 20%:16 | 20%:14 | 21%:16

How prediction computed

Window=24 Horizon=15 | AnalyzedFrames=403 | ModelReady=1

Gate=DELTA_BELOW_THRESHOLD (NOT_MET) | Threshold=3 (NOT_MET) | PredictionMod

BEGIN PREDICTION

demo clips of myproj

```
Running d:/projects/Crowd_System_AI/.venv/Scripts/python.exe train_crowd_predictor.py --videos data/crowd_videos/demo_training/*.mp4 --sample-every 4 --optimize-level aggressive -max-rows-per-series 600 --model-path models/crowd_predictor_accboost_v4.joblib --artifact-tag accboost_v4
```

```
- data/crowd_videos/demo_training/demo_clip_12.mp4
- data/crowd_videos/demo_training/demo_clip_13.mp4
data/crowd_videos/demo_training/demo_clip_01.mp4: 150
data/crowd_videos/demo_training/demo_clip_02.mp4: 150
data/crowd_videos/demo_training/demo_clip_03.mp4: 150
data/crowd_videos/demo_training/demo_clip_04.mp4: 150
```

Target Place Search

guduvanchery

Type at least 3 letters to see suggestions and click one option to jump to that location.

Calculate Capacity

Auto Zone Side (m)

19

Map Theme

Satellite

Example Inputs

1. Place Search: Guduvanchery bus stand
2. Auto Zone Side: 25 (small street block), 40 (medium block), 60 (open ground)
3. Draw rectangle/polygon exactly on usable walking area for accurate capacity.

Capacity is calculated from your marked zone only. No crowd prediction is used.

Geospatial Output

Area (m2): 148.47
Perimeter (m): 60.11
Zone Centroid: 12.837496, 80.070553
Zone Address: Guduvanchery, Chengalpattu, Tamil Nadu, 603202, India
Coordinates Used: 6

Selected Location Intelligence

Selection Source: Manual draw
Selected Latitude: 12.837558
Selected Longitude: 80.070508
Auto Zone Side: -
Updated At: 11:46:58

Capacity Estimation

Comfort Capacity	14	Safe Capacity (Recommended)	24
Upper Dense Capacity	36	Area Per Person (Safe)	0.83 m2

Recommended People Limit: 24
Usable Area: 20.03 m2 (13.5%)
Gross Reference (1x/2x/4x): 148 / 296 / 593

Capacity model: Mixed Urban context with usable-area safety deductions

Map Services

Provider: OSM | Free map stack enabled. No billing required.

Use map draw tools to draw polygon or rectangle around your target area.

Government Boys Higher Secondary School
Government Girls Higher Secondary School
Sri Vinayaga Clinic
M.M. Clinic
Animal Birth Control Centre
Government Veterinary Hospital
Jaya Clinic
Dr. Ganesh
Nandhivaram
Krishnapuram First Street
Nelikipuram Road
MDR598

Crowd Intelligence System

AI CROWD INTELLIGENCE SYSTEM

Current: 17
Future: 14
Status: SAFE

Flow: DOWN

FPS: 15

Detected Person Preview

Unique Persons: 67

ID: 65
ID: 66
ID: 67

FUTURE CROWD PREDICTION

Current: 17
Future: 14
Started at 13.9% video progress

ID 46
ID 42
ID 64
ID 54
ID 10
ID 15
ID 7
ID 40
ID 2
ID 4
ID 16
ID 37
ID 63
ID 39
ID 41
ID 53
ID 66